ReadyShell Overlay Inventory Report (v0.2)

Artifact-backed inventory for the current ReadyShell overlay system. This report is generated from the live build outputs, the ReadyShell linker map, and the active D71 disk image instead of hand-maintained notes.

Executive Summary

PROFILE	DISK SOURCE	OVERLAY WINDOW	WINDOW SIZE	RESIDENT HEAP
`precog-dual-d71`	`releases/0.2/precog-dual-d71/readyos-v0.2-dual-d71_1.d71`	`$8E00-$C5FF`	**14336 bytes**	**910 bytes**

Overlays 1-2 are boot-loaded once and cached in one shared REU bank using two full-window slots. Overlays 3-7 are loaded from disk on demand. Bank `0x48` is shared for the external-command registry, overlay metadata, pause state, command handoff scratch, and the REU-backed ReadyShell value arena.

Runtime Memory Map

- **Resident app window:** `$1000-$C5FF` (46592 bytes)
- **Overlay load bytes:** `$8DFE-$8DFF`
- **Overlay execution window:** `$8E00-$C5FF` (14336 bytes)
- **Resident BSS:** `$8878-$8A6E` (503 bytes)
- **Resident heap:** `$8A70-$8DFD` (910 bytes)
- **High RAM runtime:** `$CA00-$CFFF` (1536 bytes)

REU Layout

- **Shared cache bank:** `$400000-$40FFFF` (65536 bytes)
- **Overlay 1 parse slot:** `$400000-$4037FF` (14336 bytes)
- **Overlay 2 exec slot:** `$403800-$406FFF` (14336 bytes)
- **Shared bank free tail:** `$407000-$40FFFF` (36864 bytes)
- **Debug trace ring:** `$43F000-$43F20F` (528 bytes)
- **Command scratch:** `$480000-$487FFF` (32768 bytes)
- **Command registry header:** `$488010-$488017` (8 bytes)
- **Command descriptor table:** `$488020-$48807F` (96 bytes)
- **Overlay state table:** `$488080-$4880D9` (90 bytes)
- **Shared metadata:** `$4880E0-$4880EB` (12 bytes)
- **Pause flag:** `$4880F0` (1 byte)
- **Heap metadata:** `$488000-$4880FF`
- **Heap arena:** `$488100-$48FEFF` (32256 bytes)

Shared REU Bank Visual

Bank `0x40` : Core Overlay Cache

Overlay 1 parse slot `$400000-$4037FF`
Full `0x3800` byte window image for `rsparser.prg`. Live payload is `13005` bytes, but the cache stores the whole overlay window so writable overlay data survives phase switching.

Overlay 2 exec slot `$403800-$406FFF`
Full `0x3800` byte window image for `rsvm.prg`. Live payload is `14033` bytes.

Free tail `$407000-$40FFFF`
`36864` bytes currently left free because overlays 1 and 2 now share one bank instead of consuming separate banks.

Bank `0x48` : Shared Shell State

Command registry `$488010-$4880D9`
Registry header, external-command descriptors, and per-overlay load/cache state stored in REU metadata so new external commands do not consume resident BSS.

Shared metadata + pause `$4880E0-$4880F0`
Overlay cache metadata record plus the shared pause bit used by the resident line printer and `MORE`.

Command scratch + value arena `$480000-$48FEFF`
Command handoff scratch, REU heap metadata, and the persistent value arena for REU-backed strings, arrays, and objects.

Overlay Inventory

Ovl	Role	Build PRG	Disk Name	PRG Bytes	Disk Blocks	Live Bytes	Window Use	Commands	REU
1	Parser / Lexer	`rsparser.prg`	`rsparser`	13007	52	13005	90.7%	None directly; parse phase support.	bank `0x40` slot `$400000-$4037FF`
2	Execution Core	`rsvm.prg`	`rsvm`	14035	56	14033	97.9%	PRT, MORE, TOP, SEL, GEN, TAP and the shared execution paths that command overlays return to.	bank `0x40` slot `$403800-$406FFF`
3	Drive Info + Directory Listing	`rsdrvilst.prg`	`rsdrvilst`	7745	31	7743	54.0%	DRVI, LST	disk only
4	Load Value	`rsldv.prg`	`rsldv`	11156	44	11154	77.8%	LDV	disk only
5	Store Value	`rsstv.prg`	`rsstv`	8972	36	8970	62.6%	STV	disk only
6	File Delete / Rename / Write	`rsfops.prg`	`rsfops`	13235	53	13233	92.3%	DEL, REN, PUT, ADD	disk only
7	File Read / Copy	`rscat.prg`	`rscat`	13042	52	13040	91.0%	CAT, COPY	disk only

Command Topology

This view is command-centric rather than file-centric. Shared overlays stand out here because multiple command pills sit inside one overlay card.

Overlay 2 — SHARED EXECUTION CORE
`rsvm`
Shared execution paths that command overlays return to.
PRT · MORE · TOP · SEL · GEN · TAP

Overlay 3 — SHARED COMMAND OVERLAY
`rsdrvilst`
Shared command overlay for DRVI and LST.
DRVI · LST

Overlay 4 — SINGLE-COMMAND OVERLAY
`rsldv`
Single-command overlay for LDV.
LDV

Overlay 5 — SINGLE-COMMAND OVERLAY
`rsstv`
Single-command overlay for STV.
STV

Overlay 6 — SHARED COMMAND OVERLAY
`rsfops`
Shared command overlay for DEL, REN, PUT, and ADD.
DEL · REN · PUT · ADD

Overlay 7 — SHARED COMMAND OVERLAY
`rscat`
Shared command overlay for CAT and COPY.
CAT · COPY

Resident Program

- **Build PRG:** `readyshell.prg`
- **Disk name:** `readyshell`
- **Disk footprint:** 30842 bytes, 122 D71 blocks
- **Resident sources:** `readyshellpoc.c, rs_token.c, rs_bc.c, rs_errors.c, rs_cmd_registry.c, rs_vm_c64.c, rs_overlay_c64.c, rs_platform_c64.c, rs_screen_c64.c, tui_nav.c, reu_mgr_dma.c, resume_state_ctx.c, resume_state_core.c`
- **Resident asm/runtime:** `rs_runtime_c64.s`
- **Role:** Resident app shell loop plus vm/overlay runtime. Command tokens resolved here, then dispatched to overlay 2 or command overlays.
- **Current resident segments:** `CODE $734B`, `RODATA $0455`, `DATA $0051`, `INIT $001C`, `ONCE $0038`, `BSS $01F7`

Per-Overlay Detail

Overlay 1: Parser / Lexer

Lexer, parser, AST construction, and parse cleanup.

- **Build PRG:** `rsparser.prg`
- **Disk staging PRG:** `obj/rsparser.prg`
- **Disk filename:** `rsparser`
- **Source files:** `rs_lexer.c, rs_parse.c, rs_parse_support.c, rs_parse_free.c`
- **Commands:** None directly; parse phase support.
- **Live window range:** `$8E00-$C0CC` (13005 bytes)
- **Window share:** 90.7% used, 1331 bytes free
- **Disk footprint:** 13007 bytes, 52 D71 blocks
- **REU policy:** Boot-loaded from disk, then cached in the shared core-overlay REU bank.
- **RAM notes:** Lives entirely inside the shared overlay window while active.

Overlay 2: Execution Core

Values, variables, formatting, pipes, and shared execution helpers.

- **Build PRG:** `rsvm.prg`
- **Disk staging PRG:** `obj/rsvm.prg`
- **Disk filename:** `rsvm`
- **Source files:** `rs_vars.c, rs_value.c, rs_format.c, rs_cmd.c, rs_pipe.c`
- **Commands:** PRT, MORE, TOP, SEL, GEN, TAP and the shared execution paths that command overlays return to.
- **Live window range:** `$8E00-$C4D0` (14033 bytes)
- **Window share:** 97.9% used, 303 bytes free
- **Disk footprint:** 14035 bytes, 56 D71 blocks
- **REU policy:** Boot-loaded from disk, then cached in the shared core-overlay REU bank.
- **RAM notes:** Includes rs_vm_fmt_buf[128] and rs_vm_line_buf[384] inside the overlay image.

Overlay 3: Drive Info + Directory Listing

Shared command overlay for DRVI and LST.

- **Build PRG:** `rsdrvilst.prg`
- **Disk staging PRG:** `obj/rsdrvilst.prg`
- **Disk filename:** `rsdrvilst`
- **Source files:** `rs_cmd_lst_c64.c, rs_cmd_drvi_c64.c`
- **Commands:** DRVI, LST
- **Live window range:** `$8E00-$AC3E` (7743 bytes)
- **Window share:** 54.0% used, 6593 bytes free
- **Disk footprint:** 7745 bytes, 31 D71 blocks
- **REU policy:** Loaded from disk on demand for each command call. No dedicated REU cache.
- **RAM notes:** Shares the inter-command REU handoff area at 0x480000-0x487FFF.

Overlay 4: Load Value

Single-command overlay for LDV.

- **Build PRG:** `rsldv.prg`
- **Disk staging PRG:** `obj/rsldv.prg`
- **Disk filename:** `rsldv`
- **Source files:** `rs_cmd_ldv_c64.c`
- **Commands:** LDV
- **Live window range:** `$8E00-$B991` (11154 bytes)
- **Window share:** 77.8% used, 3182 bytes free
- **Disk footprint:** 11156 bytes, 44 D71 blocks
- **REU policy:** Loaded from disk on demand for each command call. No dedicated REU cache.
- **RAM notes:** Uses the shared handoff region plus the REU-backed value arena in bank 0x48 when hydrating pointer-backed values.

Overlay 5: Store Value

Single-command overlay for STV.

- **Build PRG:** `rsstv.prg`
- **Disk staging PRG:** `obj/rsstv.prg`
- **Disk filename:** `rsstv`
- **Source files:** `rs_cmd_stv_c64.c`
- **Commands:** STV
- **Live window range:** `$8E00-$B109` (8970 bytes)
- **Window share:** 62.6% used, 5366 bytes free
- **Disk footprint:** 8972 bytes, 36 D71 blocks
- **REU policy:** Loaded from disk on demand for each command call. No dedicated REU cache.
- **RAM notes:** Uses the shared handoff region plus the REU-backed value arena in bank 0x48 when serializing pointer-backed values.

Overlay 6: File Delete / Rename / Write

Shared command overlay for DEL, REN, PUT, and ADD.

- **Build PRG:** `rsfops.prg`
- **Disk staging PRG:** `obj/rsfops.prg`
- **Disk filename:** `rsfops`
- **Source files:** `rs_cmd_delren_c64.c, rs_cmd_putadd_c64.c`
- **Commands:** DEL, REN, PUT, ADD
- **Live window range:** `$8E00-$C1B0` (13233 bytes)
- **Window share:** 92.3% used, 1103 bytes free
- **Disk footprint:** 13235 bytes, 53 D71 blocks
- **REU policy:** Loaded from disk on demand for each command call. No dedicated REU cache.
- **RAM notes:** Keeps file-operation staging and transient command state in overlay-local code plus the shared REU scratch region.

Overlay 7: File Read / Copy

Shared command overlay for CAT and COPY.

- **Build PRG:** `rscat.prg`
- **Disk staging PRG:** `obj/rscat.prg`
- **Disk filename:** `rscat`
- **Source files:** `rs_cmd_cat_c64.c, rs_cmd_copy_c64.c`
- **Commands:** CAT, COPY
- **Live window range:** `$8E00-$C0EF` (13040 bytes)
- **Window share:** 91.0% used, 1296 bytes free
- **Disk footprint:** 13042 bytes, 52 D71 blocks
- **REU policy:** Loaded from disk on demand for each command call. No dedicated REU cache.
- **RAM notes:** Uses overlay-local file I/O logic plus shared REU scratch when line staging or copy state is needed.

Observations

- Overlay 2 is effectively full at 14033 bytes of 14336 (97.9%).
- Overlay 1 is also large at 13005 bytes (90.7%).
- The resident heap below the overlay load address is only 910 bytes, so large working sets depend on overlays and REU-backed storage.
- Overlays 1-2 now share bank `0x40` and leave a free tail of 36864 bytes in that bank instead of consuming separate banks.
- Command overlays 3-7 are smaller, but they pay a disk-load cost on every command invocation because they are not cached in fixed REU banks today.
- Overlay 2 owns the shared formatting buffers, which is why it consumes almost the entire overlay window.